

NEXUS BOARD

Multi-User Engineering Notes & Task Management System

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01 // OVERVIEW

What is NEXUS BOARD?

NEXUS BOARD is a real-time, browser-based collaborative notes and task management platform engineered for technical teams. It combines the aesthetics of cyberpunk interface design with macOS-style window chrome and circuit-board visual motifs, creating an environment suited to electrical, electronics, communications, and information engineering workflows.

The system supports multiple simultaneous users, each identified by a personal name or handle attached to every note as a signature badge. Notes persist across browser sessions using local storage, ensuring continuity between work cycles.

Core Capabilities

Signal Capture	Create structured notes in four distinct formats: Note, To-Do, Checklist, and Alert.
Checklist Engine	Interactive, toggleable checklist items with real-time progress tracking.
Multi-User Identity	Every note carries an author signature with auto-coloured avatar initials.
Persistent Storage	Notes are saved to local storage and survive page refreshes or browser restarts.
Filter System	Instantly filter the board by note type to reduce cognitive load.
Live Clock	Real-time system clock displayed in the macOS-style top bar.

02 // GETTING STARTED

System Requirements

NEXUS BOARD runs entirely in a modern web browser with no installation required. The following environments are supported:

COMPONENT	REQUIREMENT	NOTES
Browser	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+	JavaScript must be enabled
Screen Width	Minimum 480px	Optimal at 900px+
Network	Not required after initial load	Fully offline-capable
Storage	LocalStorage enabled	For note persistence
Fonts	Google Fonts (JetBrains Mono)	Falls back to system monospace

Quick-Start in 3 Steps

STEP 01 Open the Application

Load the NEXUS BOARD HTML file in your browser. The board initialises automatically with sample notes demonstrating each note type.

STEP 02 Set Your Identity

Type your name or engineering handle (e.g. "Priya.K", "Dev.T") into the "Your name / handle" field in the input panel at the top. This becomes your signature on every note you create.

STEP 03 Post Your First Note

Enter a title, choose a note type, write your content, then click TRANSMIT NOTE. Your note appears instantly on the board.

03 // INTERFACE LAYOUT

Application Zones

NEXUS BOARD is divided into four distinct interface zones, each serving a specific function in the engineering workflow:

ZONE A	Top Bar	Contains the macOS-style window control dots (red / yellow / green), the application name and version tag, and a live real-time clock. This bar uses frosted-glass blur over the circuit-board background grid.
ZONE B	Input Panel	The primary data-entry panel. Contains fields for author name, note title, note type selector, main content area, and (conditionally) the checklist item entry field. A gradient top border indicates active input mode.
ZONE C	Filter Bar	A row of toggle buttons allowing instant filtering of the notes grid by type: ALL, NOTES, TO-DO, CHECKLIST, ALERTS. Active filter is highlighted in cyan.
ZONE D	Notes Grid	The main content area. Notes are displayed as cards in a responsive auto-fill grid. Each card shows a type accent stripe, title, content, type badge, body content or checklist, author avatar, and timestamp.
ZONE E	Status Bar	A slim footer bar showing system status, total signal count, unique user count, and storage mode indicator.

Visual Design Language

The visual identity of NEXUS BOARD draws from multiple engineering aesthetics unified into a single coherent system:

INFLUENCE	VISUAL ELEMENT
Cyberpunk	Cyan glow borders, dark panel overlays, monospace typography throughout
PCB / Electronics	Circuit-board background grid, trace-like colour gradients, "signal" terminology
macOS	Traffic-light window dots (red/yellow/green), frosted-glass topbar, card corner radii
Mechanical	Precision layout grid, status indicators, version/revision labelling
Information Eng.	Type badges, filter system, author signatures, timestamp metadata

04 // NOTE TYPES

Signal Types Reference

NEXUS BOARD supports four note types (referred to as "signal types" in the interface). Each type has a distinct visual identity and intended use case:

■ NOTE

General-purpose information capture. Use for observations, meeting notes, protocol results, configuration details, or any free-form content that does not require action tracking.

■ TO-DO

A single action item or task. Use for items assigned to a team member that require completion. To-Do cards display a cyan accent stripe and badge.

■ CHECKLIST

A structured list of sub-tasks, verification steps, or review criteria. Each item can be individually ticked off. A progress bar shows overall completion percentage.

■ ALERT

A high-visibility warning or critical system note. Use for hardware faults, power rail issues, safety concerns, or items requiring immediate engineering attention.

! CAUTION

The Alert type is intended for engineering-critical issues. Posts of this type are visually prominent and should be resolved or acknowledged promptly.

05 // CREATING NOTES

The Input Panel

FIELD	TYPE	REQUIRED	DESCRIPTION
Your name / handle	Text	No*	Author identity. Displayed as avatar initials + name on the card. Default is your browser name.
Note title	Text (max 60)	YES	The headline of your note. Displayed prominently in the card header.
Type selector	Dropdown	YES	Selects the note type: Note, To-Do, Checklist, or Alert. Changing to another type will clear the content area.
Content area	Textarea	No	Main body of the note. Accepts free-form text. Supports multi-line input.
Checklist items	Textarea	Conditional	Visible only when type is set to Checklist. Enter one item per line.

> TIP

The author name field is not required but strongly recommended for multi-user boards. Your handle persists in the field between notes – type it once per session.

Submitting a Note

Click the TRANSMIT NOTE button (or press Tab to reach it and Enter to activate). The note is immediately added to the top of the board and saved to local storage. The input fields are cleared automatically (except the author name) to allow rapid sequential posting.

Deleting a Note

Click the small X button in the top-right corner of any note card. Deletion is immediate and permanent – there is no undo. The note is removed from both the display and local storage.

06 // CHECKLIST FEATURES

Using Checklists

Checklists are NEXUS BOARD's most powerful feature for engineering verification workflows, pre-flight checks, test protocols, and sprint reviews.

Creating a Checklist

01. Set the type dropdown to "■ Checklist".
02. A second textarea labelled "Checklist items" appears below the content field.
03. Enter each checklist item on its own line.
04. Add a title and optional description in the content field.
05. Click TRANSMIT NOTE to publish.

Interacting with Checklist Items

Each checklist item on a published card is clickable. Clicking an item toggles it between incomplete and complete states:

STATE	VISUAL	MEANING
Incomplete	Empty square box, normal text colour	Item has not yet been actioned
Complete	Amber-filled box with checkmark, strikethrough text	Item has been verified or actioned

Progress Bar

Every checklist card displays a slim amber progress bar immediately above the item list. The bar fills proportionally as items are completed. The calculation is:

$$\text{Progress (\%)} = (\text{Completed Items} / \text{Total Items}) \times 100$$

The progress bar animates smoothly on state change and is updated in real-time without any page reload.

>> NOTE

Checklist state (checked/unchecked) is persisted to local storage immediately on each toggle, so progress is preserved across page refreshes.

07 // MULTI-USER COLLABORATION

User Identity System

NEXUS BOARD is designed for shared use by engineering teams. Each note carries a permanent author signature set at creation time. The identity system works as follows:

Handle / Name	Entered in the "Your name / handle" field. Can be a real name (e.g. "Priya K"), a username (e.g. "dev.t"), or an engineering callsign.
Avatar Initials	The first letter of each word in the name is extracted and displayed in a circular avatar. "Priya K" becomes "PK".
Avatar Colour	Colour is deterministically derived from the name string using a hash function. The same name always produces the same colour across all sessions and devices.
Timestamp	A creation timestamp (time + date) is recorded and displayed in the card footer. This uses the local system time of the posting user.

! CAUTION

NEXUS BOARD does not enforce unique usernames. Any text entered in the name field is accepted. Teams should agree on a naming convention (e.g. "Firstname.LastInitial") to keep

08 // FILTERING & NAVIGATION

Filter System

The filter bar sits between the input panel and the notes grid. It provides one-click filtering of visible notes by type:

FILTER BUTTON	SHOWS
ALL	All notes regardless of type (default state)
■ NOTES	Only general-purpose Note cards
■ TO-DO	Only To-Do action item cards
■ CHECKLIST	Only Checklist cards (with progress bars)
■ ALERTS	Only Alert cards – useful for rapid triage

Filtering is non-destructive: notes are hidden, not deleted. Switching filters or returning to ALL restores all notes immediately. The status bar always reflects the total note count and user count regardless of active filter.

09 // DATA PERSISTENCE

How Notes Are Saved

NEXUS BOARD uses the browser's built-in `localStorage` API to persist notes between sessions. All data is stored locally on the user's device – no server, no network, no account required.

BEHAVIOUR	DETAIL
Auto-save on post	Notes are saved immediately when TRANSMIT NOTE is clicked
Auto-save on check	Checklist item toggles are saved instantly
Auto-save on delete	Note removal is persisted immediately
Page refresh	All notes are fully restored on reload
Browser restart	Notes persist as long as <code>localStorage</code> is not cleared
Cross-tab	Notes created in one tab are not automatically reflected in other open tabs
Storage key	Data stored under key: <code>nexus_notes_v1</code>
Data format	JSON array of note objects serialised as a string

! CAUTION

Clearing browser history, site data, or `localStorage` will erase all notes. For critical data, export your notes by copying the `localStorage` value from browser DevTools before

10 // KEYBOARD & UX TIPS

Productivity Tips

■ Name once per session

Enter your author handle at the start of your session. It remains in the field, so all subsequent notes carry your signature automatically.

■ Use Tab to navigate

Tab through the input fields: Name → Title → Type → Content → (Checklist items) → TRANSMIT. This keeps your hands on the keyboard.

■ Alert for blockers

Use the Alert type for anything that blocks engineering progress. The red visual makes blockers immediately visible on the shared board.

■ Checklists for protocols

Convert test protocols, PCB review checklists, and deployment runbooks into Checklist notes. The progress bar gives instant status.

■ Short titles, long content

Keep titles short and descriptive (like a commit message). Put detail in the content area.

■ Filter during triage

During standups or reviews, use the ALERTS filter to quickly surface critical items, then switch to TO-DO for task review.

11 // TROUBLESHOOTING

Common Issues

■ Notes not appearing after refresh

Cause: `localStorage` may be disabled or in private/incognito mode.

Fix: Enable `localStorage` in browser settings, or use standard (non-private) browsing mode.

■ Checklist item textarea not showing

Cause: The type dropdown is not set to "■ Checklist".

Fix: Select "■ Checklist" from the type dropdown – the textarea appears automatically.

■ Avatar shows wrong initials

Cause: Name contains non-alphabetic characters or is very short.

Fix: Use a name with at least two words or two characters (e.g. "A.B" or "Dev Team").

■ Font renders as plain monospace

Cause: Google Fonts (JetBrains Mono) could not load – likely no internet connection.

Fix: The interface falls back to the system monospace font. Functionality is unaffected.

■ Board is empty on load

Cause: First launch (no stored data) or `localStorage` was cleared.

Fix: This is expected on first launch. Default sample notes load automatically. If they don't appear, refresh the page.

■ Notes from another user not visible

Cause: `localStorage` is device-local. Notes from other users on other devices are not synchronised automatically.

Fix: For true multi-user sync, a shared backend or cloud storage integration is required (not included in the base version).

12 // TECHNICAL REFERENCE

Data Schema

Each note is stored as a JSON object with the following structure:

```
{
  "id": Number — Unix timestamp of creation (unique ID)
  "type": String — "note" | "todo" | "checklist" | "alert"
  "title": String — Note headline (max 60 chars)
  "content": String — Main body text
  "author": String — Author name or handle
  "ts": Number — Unix timestamp (milliseconds)
  "items": Array — Checklist items (empty array for other types)
}
```

Checklist item object:

```
{
  "text": String — Item label
  "done": Boolean — Completion state
}
```

Avatar Colour Algorithm

Avatar background colour is derived deterministically from the author string using a polynomial hash function mapped to a fixed palette of 8 colours:

```
PALETTE = [
  #00B8D9, #7C3AED, #FF5F57, #00C853,
  #FF9500, #E91E8C, #0070F3, #8B5CF6
]

hash(name):
  h = 0
  for each character c in name:
    h = (h * 31 + charCode(c)) mod 8
  return PALETTE[h]
```

Technology Stack

LAYER	TECHNOLOGY	PURPOSE
Rendering	Vanilla HTML5 / CSS3	DOM structure and styling
Logic	Vanilla JavaScript (ES6+)	State management, events, storage
Typography	JetBrains Mono (Google Fonts)	Monospace engineering aesthetic
Icons	Tabler Icons (outline set)	UI action icons
Storage	localStorage API	Client-side note persistence

Networking

None

Fully offline – no external dependencies at runtime